Huawei Access Node JAY-ACC-77 Failure Resolution SOP

Node: JAY-ACC-77 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_7768

Description

The JAY-ACC-77 site, utilizing Huawei's Access equipment, is prone to hardware/environmental faults that manifest as ASIC memory parity errors, line card crashes, or power supply redundancy loss. This is due to the inherent design of the hardware components, specifically the Line Cards that employ ASICs to perform complex processing tasks. When these components fail, the system's ability to maintain data integrity and provide reliable services is compromised, resulting in the ALARM_7768 alarm being triggered. Furthermore, the redundancy mechanism, designed to ensure continuity of service, can be compromised if a power supply failure occurs, necessitating immediate attention.

Pre-requisite Checks

- check: Verify redundant link status

- check: Ensure remote hands are dispatched to the Jayanagar Metro Site

- check: Confirm power supply redundancy is intact

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

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Command: display fault brief, Expected Output: Fault Brief information for Line Card 0/0/1, Description: This command displays a brief summary of the faults detected on the system, specifically for the Line Card 0/0/1. It helps identify the root cause of the issue.

Command: display power supply status, Expected Output: Power Supply Status: Redundant Power Supply Module 1 is not available, Description: This command displays the status of the power supply modules, highlighting any redundancy issues. It helps verify the power supply redundancy.
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Resolution Steps

Step 1: Step: Replace the faulty Line Card, Command: reset line-card 0/0/1, Description: This command resets the Line Card 0/0/1, which helps restore the system's functionality. After resetting, the

system may still be in a failed state and may require further action to resolve the issue.

Step 2: Step: Restore power supply redundancy, Command: install power supply module 1, Description: This command installs a new power supply module (PSM) to restore redundancy. This ensures the system can maintain its services and continue to operate reliably.

Step 3: Step: Reboot the node, Command: reset system, Description: This command reboots the entire node, ensuring all components are restarted and the system is restored to its operational state.

Post-Check Verification

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Command: display system state, Expected Output: System State: Online, Description: This command verifies the system's state, confirming it is online and operational.

Command: display alarm information, Expected Output: Alarm Information: ALARM_7768 cleared, Description: This command verifies the alarm status, confirming the ALARM_7768 alarm has been cleared.
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Roll-back Procedures

- Step: Restore the original power supply module, Command: remove power supply module 1, Description: This command removes the new power supply module, replacing it with the original module to restore the system's original state.

- Step: Reboot the node (again), Command: reset system, Description: This command reboots the node to ensure all components are restarted and the system is restored to its original state after the rollback.